# 5. WIDE AREA NETWORKS

# Course: Diploma in Cyber Security

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## Introduction

A WAN connects networks over a large geographical area. The internet is the most common example. Businesses can also use leased lines, MPLS and other WAN services to connect offices in different cities.

## Task 1: Phone to Instagram WAN diagram

Show the phone connected to home Wi-Fi, then ISP equipment and at least two WAN routers/switches before reaching the Instagram server. The WAN section should be clearly labelled.

**[EVIDENCE / NOTE: Phone → Home Router → ISP → WAN Router 1 → WAN Router 2 → Internet → Server.]**

**Figure: Smartphone to Instagram Server WAN Data Path**

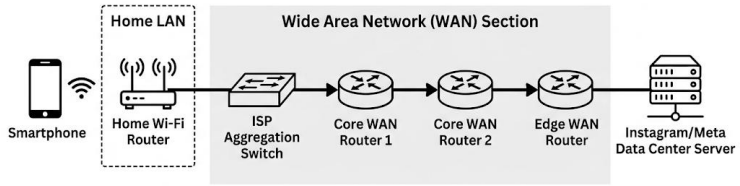



**Figure 1: End-to-End WAN Connectivity Path from Home Client to Instagram Edge Infrastructure**

# Task 2: Compare WAN technologies

MPLS, leased lines and broadband can be compared by speed, cost, reliability and use case.

| Technology | Bandwidth / Speed | Typical Cost | Reliability & SLA | Primary Use Case |
|---|---|---|---|---|
| **MPLS (Multiprotocol Label Switching)** | Scalable (10 Mbps – 10 Gbps) | Higher | High (Strict carrier SLAs, QoS packet prioritization) | Connecting enterprise multi-branch offices requiring low-latency VoIP and ERP systems. |
| **Leased Line (Dedicated T1/E1, Dark Fiber)** | Fixed dedicated symmetric speeds (up to 10+ Gbps) | High (Highest recurring cost) | Maximum (99.99% uptime, private unshared bandwidth) | Mission-critical datacenter-to-datacenter links and direct financial trading backbones. |
| **Broadband Internet (Fiber/Cable/DSL)** | High asymmetric download (up to 1 Gbps), variable upload | Lower | Best-effort (No strict latency guarantees, shared medium) | Residential use, small branch offices, and secondary backup/SD-WAN underlay links. |

# Task 3: VPN over WAN

A VPN creates an encrypted tunnel over a public network. At a coffee shop, the laptop first connects to public Wi-Fi, then the VPN client creates a secure tunnel to the VPN server before accessing online services.

Figure: Remote Access VPN Tunnel over Public WAN

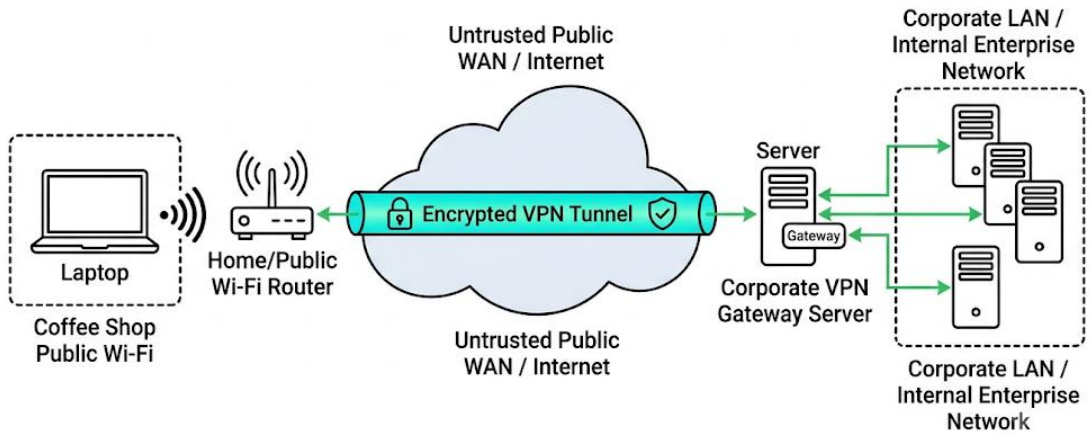



**Figure 2: Remote Access VPN Encrypted Tunnel Bypassing Insecure Public WAN Nodes**

# Task 4: Connecting Flipkart warehouses

For a large company connecting warehouses in Ahmedabad, Mumbai and Delhi, a business WAN service such as MPLS or another managed private WAN can be considered. The final choice depends on required reliability, cost, application needs and provider availability.

| Technology | Typical Cost | Use |
|---|---|---|
| MPLS | Higher | Business branch networks |
| Leased Line | High | Dedicated office-to-office links |
| Broadband Internet | Lower | Homes and small offices |

# Conclusion

This topic helped me understand the basic concept and how it can be used in a real network. I also understood that practical results should be checked on the actual system or simulator instead of assuming the result.